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Executive Summary

This project analyzes over 100,000 taxi and rideshare trips to support the city’s goal of reducing traffic congestion and improving public transportation efficiency. The dataset includes trip details such as timestamps, GPS locations, zone identifiers, and trip duration, along with integrated weather data to understand external influences on mobility. The study aims to extract meaningful patterns in urban travel behavior across time, space, and environmental conditions. The results are intended to support data-driven urban planning and transport optimization. Based on the analysis, the study provides recommendations to improve public transport planning, optimize traffic flow in high-demand areas, and incorporate weather-aware decision-making in city transport systems.

Data Understanding and Preprocessing

The taxi dataset is based on the 2016 NYC Yellow Cab trip records provided by the NYC Taxi and Limousine Commission (TLC) via Google BigQuery, split into training and testing sets. It is combined with a 366-day weather dataset containing temperature, precipitation, snowfall, and snow depth to analyze environmental effects on trip patterns.

id	Unique identifier for each trip	Nominal	Discrete
vendor_id	Code indicating the provider	Nominal	Discrete
pickup_datetime	Date and time when the trip started	Interval	Continuous
dropoff_datetime	Date and time when the trip ended	Interval	Continuous
passenger_count	Number of passengers in the vehicle	Ratio	Discrete
pickup_longitude	Longitude of pickup location	Ratio	Continuous
pickup_latitude	Latitude of pickup location	Ratio	Continuous
dropoff_longitude	Longitude of dropoff location	Ratio	Continuous
dropoff_latitude	Latitude of dropoff location	Ratio	Continuous
store_and_fwd_flag	Flag if trip was stored in vehicle memory before sending	Nominal	Discrete
trip_duration	Duration of the trip in seconds	Ratio	Continuous

	vendor_id	passenger_count	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude	trip_duration
count	1.458644e+06	1.458644e+06	1.458644e+06	1.458644e+06	1.458644e+06	1.458644e+06	1.458644e+06
mean	1.534950e+00	1.664530e+00	-7.397349e+01	4.075092e+01	-7.397342e+01	4.075180e+01	9.594923e+02
std	4.987772e-01	1.314242e+00	7.090186e-02	3.288119e-02	7.064327e-02	3.589056e-02	5.237432e+03
min	1.000000e+00	0.000000e+00	-1.219333e+02	3.435970e+01	-1.219333e+02	3.218114e+01	1.000000e+00
25%	1.000000e+00	1.000000e+00	-7.399187e+01	4.073735e+01	-7.399133e+01	4.073588e+01	3.970000e+02
50%	2.000000e+00	1.000000e+00	-7.398174e+01	4.075410e+01	-7.397975e+01	4.075452e+01	6.620000e+02
75%	2.000000e+00	2.000000e+00	-7.396733e+01	4.076836e+01	-7.396301e+01	4.076981e+01	1.075000e+03
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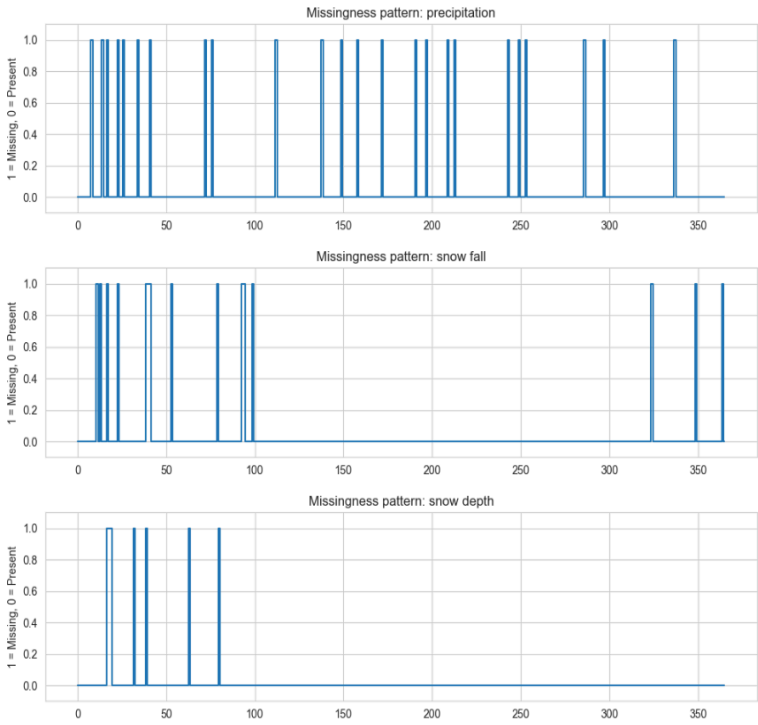
The data set has 1,458,644 entries and 11 features and has the following types of features:

- 4 Numerical Floating Point Features: These are the GPS coordinates for pickup and drop-off points.
- 3 Integer Features: vendor id, passenger count, and target trip duration.
- 4 String-Based Features: These are the unique identifiers, flags, and timestamps.
- No invalid trips were found where drop-off occurs before pickup, ensuring data consistency.
- Pickup dates range from 2016-01-01 to 2016-06-30, while drop-off dates extend to 2016-07-01.
- The dataset contains no missing or duplicate rows and includes two vendors (1 and 2).
- Passenger count anomalies include 60 zero-passenger trips and 5 high-occupancy trips (>6 passengers).
- Extreme trip durations are present, including 8,695 ultra-short trips (<1 minute) and 4 ultra-long trips (>1 day).
- Spatial inconsistencies exist with 119–805 trips outside NYC coordinate ranges and 5,897 zero-distance trips

Possible data inconsistencies

- Passenger count anomalies include 60 zero-passenger trips and 5 high-occupancy trips (>6 passengers).
- Extreme trip durations are present, including 8,695 ultra-short trips (<1 minute) and 4 ultra-long trips (>1 day).
- Spatial inconsistencies exist with 119–805 trips outside NYC coordinate ranges and 5,897 zero-distance trips
- Short-distance trips (<0.2 km) and trips with average speed greater than 150 km/h were identified.

The weather dataset contains no missing rows and matches the expected full-year range (2016-01-01 to 2016-12-31 with 366 records).No duplicate records are present; however, placeholder or missing values exist in some fields (precipitation: 24 ‘T’ values, snow fall: 15 ‘T’, snow depth: 7 ‘T’, and ‘M’ appears only as 0 cases).



Missing values in the weather dataset were filled using time-based **interpolation** for numeric features, which estimates values based on adjacent time-ordered observations. Any remaining missing entries were then handled using forward fill and backward fill to ensure a complete and consistent dataset.

	maximum temperature	minimum temperature	average temperature	precipitation	snow fall	snow depth
count	366.000000	366.000000	366.000000	366.000000	366.000000	366.000000
mean	64.625683	49.806011	57.215847	0.131148	0.137295	0.278689
std	18.041787	16.570747	17.124760	0.321028	1.606199	1.928569
min	15.000000	-1.000000	7.000000	0.000000	0.000000	0.000000
25%	50.000000	37.250000	44.000000	0.000000	0.000000	0.000000
50%	64.500000	48.000000	55.750000	0.000000	0.000000	0.000000
75%	81.000000	65.000000	73.500000	0.067500	0.000000	0.000000
max	96.000000	81.000000	88.500000	2.310000	27.300000	22.000000

Data Cleaning

The dataset was cleaned by removing inconsistent and unrealistic records while retaining valid trip behavior. All key numeric fields were converted properly, and missing values were dropped. Trips with zero passengers, very short duration (<60 seconds), and extremely long duration (>24 hours) were removed. GPS coordinates were filtered to stay within NYC boundaries to ensure spatial validity. Zero-distance trips with very short duration were treated as invalid and removed, while higher passenger counts (>6) were retained as they can be realistic.

Temporal features were extracted from pickup and dropoff timestamps after converting them into proper datetime format. From these timestamps, multiple time-based variables were derived including hour, day of the month, month, and day of the week for both pickup and dropoff events. In addition, time of day categorical labels were generated based on the hour of travel, dividing the day into meaningful segments such as Night, Morning, Afternoon, Evening, and Late Night to capture variations in travel demand across different periods.

Additional behavioral indicators were engineered including a weekend flag to identify Saturday and Sunday trips and a cross day trip indicator to capture trips that start and end on different calendar days. These features help in understanding temporal patterns and anomalies in travel behavior.

The weather dataset was then integrated with the trip data. Weather column names were cleaned for consistency and date fields in both datasets were converted to a normalized datetime format. The weather dataset was merged twice once with pickup dates and once with dropoff dates ensuring that both start and end trip conditions were captured. After merging, redundant key columns were removed to avoid duplication and maintain dataset clarity.

Spatial features were also engineered using latitude and longitude values. The Haversine formula was applied to compute the great circle distance between pickup and dropoff locations resulting in a new feature called trip distance km. Using this, average speed km per hour was calculated by dividing distance by trip duration.

To improve data quality, extremely short distance trips less than 0.2 km were identified and removed as they likely represent GPS noise cancellations or recording errors. Similarly trips with unrealistic average speeds greater than 150 km per hour were filtered out as they are physically implausible.

After all preprocessing feature engineering merging and cleaning steps the final dataset contains 1,445,172 records and 35 features making it fully consistent and suitable for further modeling and analysis.

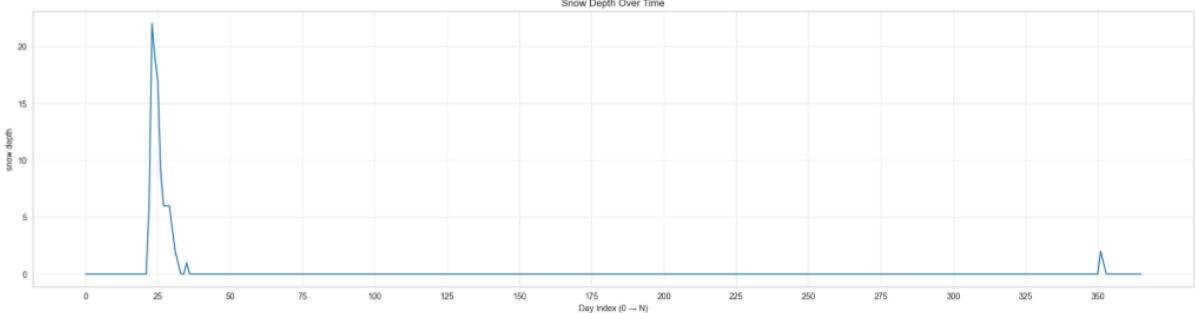
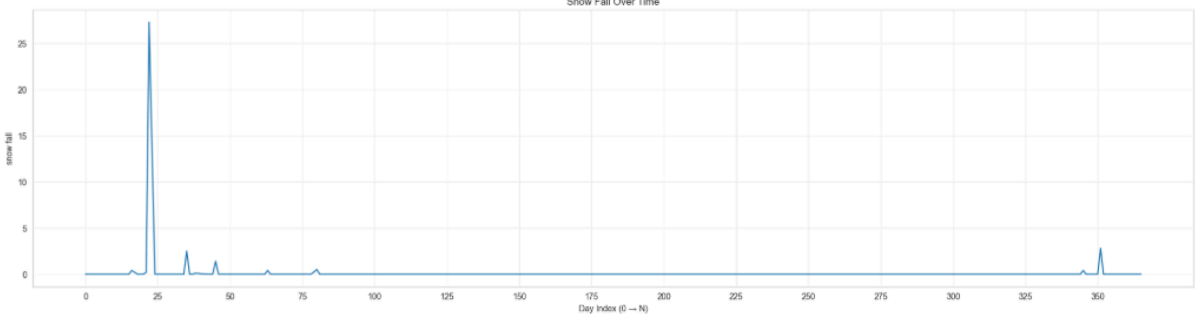
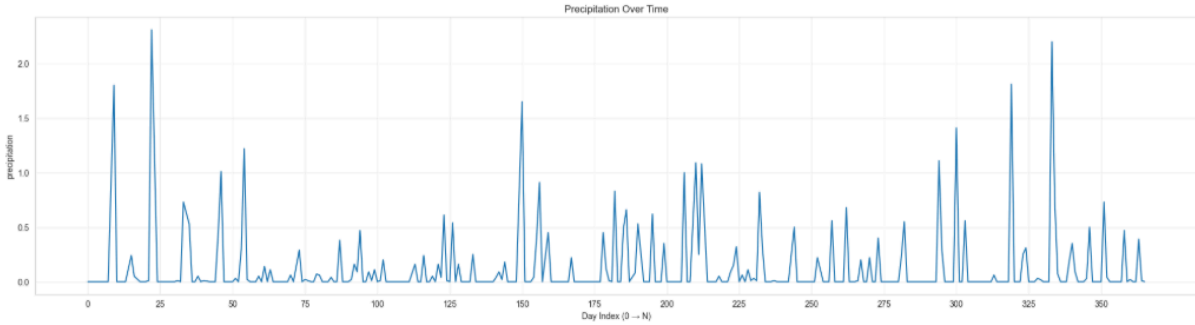
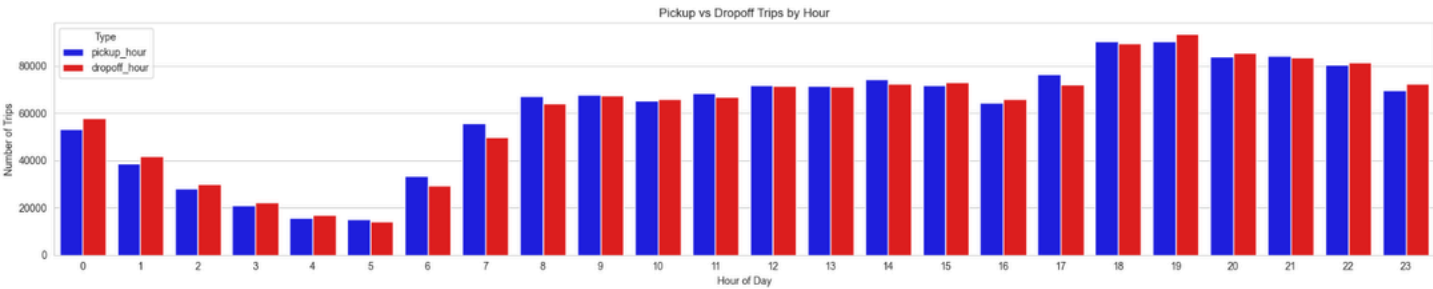
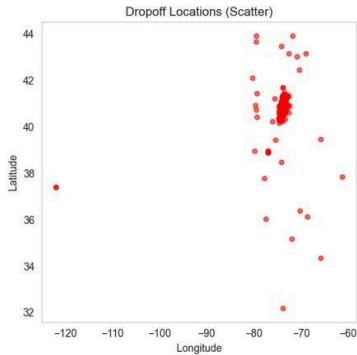
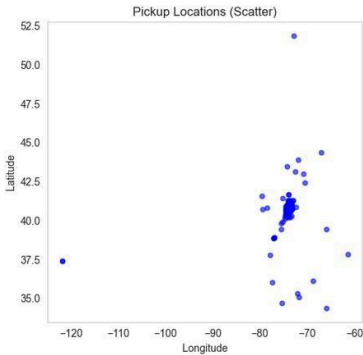
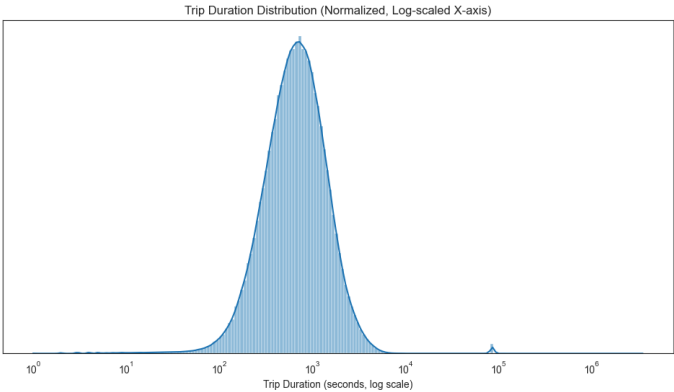
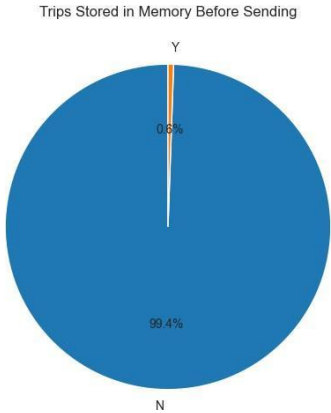
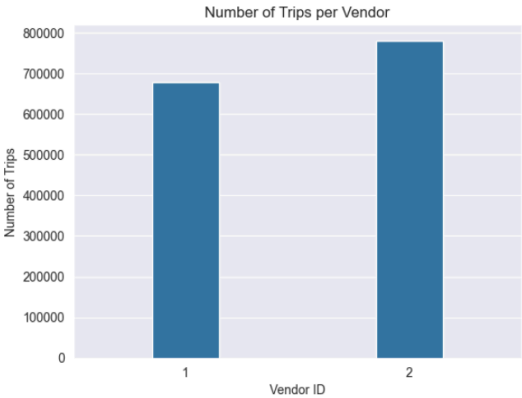
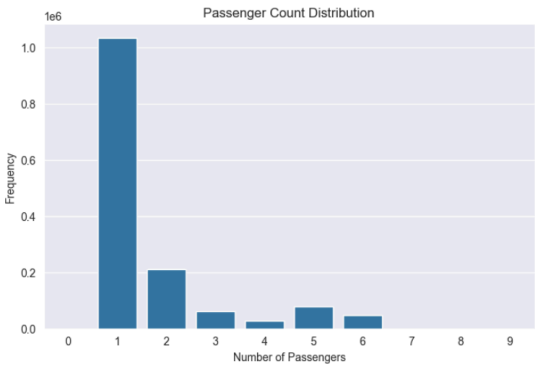
Missing values

To understand the missing data mechanism, binary missingness indicators were created for precipitation, snowfall, and snow depth, and their distributions were analyzed across temperature bins and related weather conditions. For precipitation, missing values are almost uniformly distributed across all temperature ranges with only minor variation, indicating no strong dependency on observed variables. Hence, precipitation missingness is classified as Missing Completely at Random (MCAR).

In contrast, snowfall missingness shows a clear dependency on specific temperature ranges, with a higher concentration of missing values in moderate temperature bands. This indicates that missingness is influenced by observed variables (temperature), and is therefore classified as Missing At Random (MAR).

Similarly, snow depth missingness varies with both temperature and prior extreme snow conditions, showing a structured dependence on observed environmental factors. This confirms a MAR mechanism for snow depth as well.

Overall, none of the variables show evidence of dependence on unobserved values, so there is no indication of Missing Not At Random (MNAR) behavior in the dataset.



Business proposal

Task 1: Peak-Time Traffic Management

Using Classification, predict whether a given time and location will experience congestion or not using temporal, spatial, and weather-based features. This enables proactive traffic control through signal adjustments, improved public transport scheduling, and congestion reduction during peak hours.

Task 2: Spatial Demand Region Identification

Clustering trips based on pickup locations to identify spatial demand regions across the city. This helps identify high-demand areas for taxis, enabling better placement of pickup points, improved urban planning, and optimized transport resource allocation.

Task 3: Trip Duration Prediction

Predict trip duration using spatial, temporal, and weather features and compare Linear Regression and Random Forest models. This can help improve ETA accuracy, enhance driver allocation efficiency, and support better service and pricing decisions under varying traffic and weather conditions.

Methodology

Task 1: Peak-Time Traffic Management

Features : A subset of relevant features was selected from the processed dataset, including pickup hour, pickup day of week, pickup month, weekend indicator, pickup latitude, pickup longitude, precipitation, snow fall, snow depth, average temperature, trip distance, and average speed. These features capture both temporal patterns and environmental conditions influencing traffic behavior

Target Variable Construction: Congestion was defined using a data-driven threshold based on vehicle speed. The median average speed was computed as 12.8565 km/h, which was used as the decision boundary. Trips with average speed below this value were labeled as congested (1), while those above were labeled as not congested (0).

Data Splitting Strategy: The dataset was split into training, validation, and testing sets using a 70% training, 15% validation, and 15% testing split. This ensured proper model evaluation and prevented data leakage. The feature matrix (X) included only temporal and spatial variables, while the target variable (y) represented congestion status.

Model Selection and Training: A Decision Tree Classifier was used for prediction due to its interpretability and ability to handle non-linear relationships between features. The model was trained using the training set and evaluated on the validation set for tuning purposes.

Hyperparameter Optimization: To improve model performance, Grid Search (ParameterGrid from scikit-learn) was applied. Multiple combinations of hyperparameters were trained and tested on validation data, including: max_depth, min_samples_split, min_samples_leaf, criterion (gini, entropy)

Evaluation Metrics; Model performance was assessed using multiple classification metrics:

- Accuracy (overall correctness of predictions)
- Precision (correctness of predicted congestion cases)
- Recall (ability to detect actual congestion cases)
- F1-score (balance between precision and recall)

Task 2: Spatial Demand Region Identification

Features: The clustering model was built using only spatial features, specifically pickup latitude and pickup longitude. These features represent the geographic movement patterns of taxi demand across the city. A random sample of 1,000,000 records was used from the full dataset to make computation feasible while preserving spatial distribution.

Clustering Objective: The objective of this task is to identify spatial demand regions by grouping taxi pickup locations into clusters. These clusters represent areas of similar pickup density, helping to reveal natural high-demand zones within the city.

Data Preparation Strategy: Only spatial coordinates were retained for clustering. A large-scale random sampling approach (1 million records) was used to reduce computational cost while maintaining representativeness of spatial patterns.

Model Selection and Training: K-Means Clustering was used as the primary algorithm due to its efficiency and scalability for large datasets. The model was trained for multiple values of K ranging from 2 to 11 to determine the optimal number of clusters. DBSCAN was initially considered; however, it was not used due to high computational cost on large-scale data and inconsistent performance in handling varying density regions. Therefore, K-Means was selected as the final algorithm due to its scalability and stable clustering behavior on large spatial datasets.

Evaluation Metrics:

- WCSS (Elbow Method for compactness)
- Silhouette Score (cluster quality and separation)
- Visual spatial coherence of clusters

Task 3: Trip Duration Prediction and Model Comparison

Features: A comprehensive set of spatial, temporal, weather, and operational features was selected from the processed dataset, including vendor ID, passenger count, pickup and dropoff coordinates, time-based features (hour, day, month, day of week, weekend indicator, cross-day trip flag), weather attributes (temperature, precipitation, snow fall, snow depth for both pickup and dropoff), and trip distance. These features collectively capture travel behavior, environmental conditions, and spatial movement patterns affecting trip duration.

Target Variable: Trip duration was used as the continuous target variable representing total travel time between pickup and dropoff locations.

Data Preprocessing and Transformation: The feature set was standardized before modeling. Additionally, Principal Component Analysis (PCA) was applied to reduce dimensionality while preserving variance, and a comparison was made between PCA-transformed features and the original feature space to evaluate its impact on model performance.

Data Splitting Strategy: The dataset was split into training, validation, and testing sets using a 70%–15%–15% split to ensure unbiased evaluation and prevent data leakage.

Model Selection - Linear Regression: A Linear Regression model was used as the baseline regression approach. Two variants were tested: one trained on PCA-transformed features and another on the original feature space. This allowed comparison of the impact of dimensionality reduction on predictive performance.

Model Selection - Random Forest Regressor: A Random Forest Regressor was implemented to capture non-linear relationships in the data. The model was trained using grid search-based hyperparameter tuning.

Hyperparameter Optimization: Grid search was performed over multiple configurations including: `n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`.

Evaluation Metrics: Final model performance was assessed using:

- MAE (prediction error magnitude)
- RMSE (penalized error for large deviations)
- R² Score (variance explained by the model)

Results and Interpretations

Task 1: Peak-Time Traffic Management

The final Decision Tree model was selected using entropy criterion, max depth = 10, minimum samples per leaf = 10, and minimum samples split = 20, based on validation performance.

The model achieved the following performance:

- Accuracy: 0.6908
- Precision: 0.6719
- Recall: 0.7484
- F1-score: 0.7081

The model shows balanced performance, with a relatively strong F1-score indicating good trade-off between precision and recall. Recall is higher than precision, meaning the model is better at correctly identifying congested situations, even if it slightly over-predicts congestion. Overall accuracy of ~69% indicates that congestion patterns are moderately predictable using temporal, spatial, and weather features.

Patterns Observed

- Congestion is time-based, with low levels in early morning and high levels during daytime and peak commuting hours.
- Weekdays consistently exhibit higher congestion compared to weekends, indicating work-driven travel demand.
- Congestion varies significantly across geographic zones, confirming that traffic intensity is location-dependent rather than uniform across the city.
- Weather variables such as precipitation and snow are associated with moderate changes in congestion levels, indicating environmental influence.
- Interaction between time and day type show that weekday peak hours are significantly more congested than weekend peak hours.

Interpretation

Traffic congestion is not random but follows clear spatio-temporal and environmental patterns. The model captures these relationships effectively, showing that congestion is driven by predictable commuting behavior combined with location-specific demand and weather conditions.

Actionable Recommendations

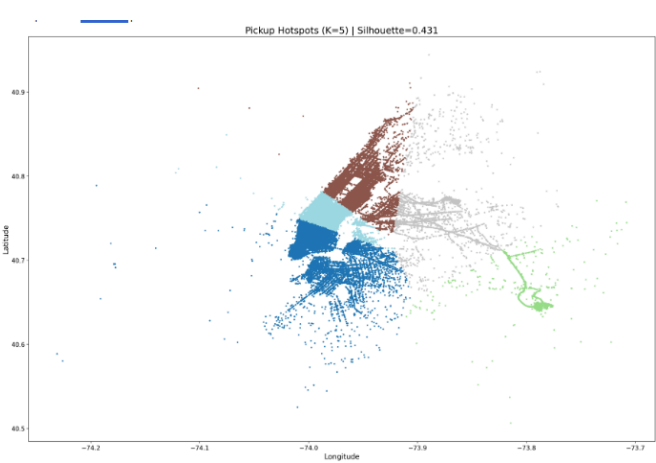
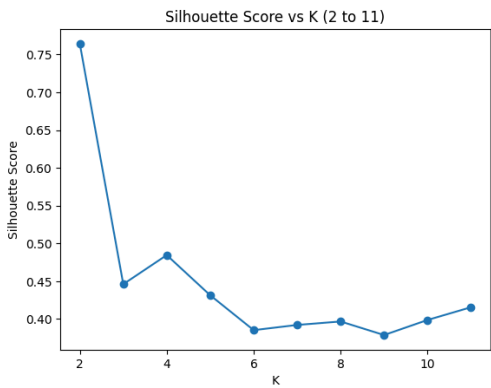
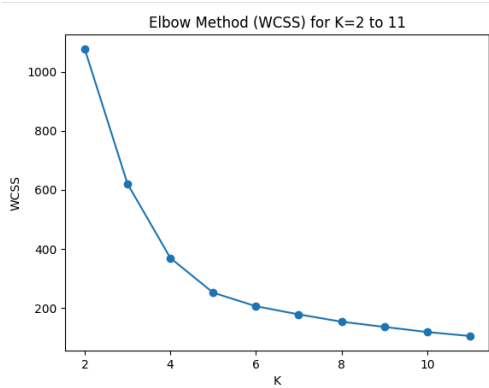
- The congestion prediction model can be used to build a real-time traffic warning system that identifies high-risk zones in advance and helps authorities reroute traffic and adjust signals before congestion builds up.
- Traffic signal systems can be made adaptive by using predicted congestion levels to automatically change signal timings at busy intersections, improving traffic flow in areas expected to become crowded.
- Public transport services such as buses and taxis can be scheduled more efficiently by using congestion forecasts, ensuring higher availability during expected peak hours and reducing the dependence on private vehicles in busy zones.

Task 2: Spatial Demand Region Identification

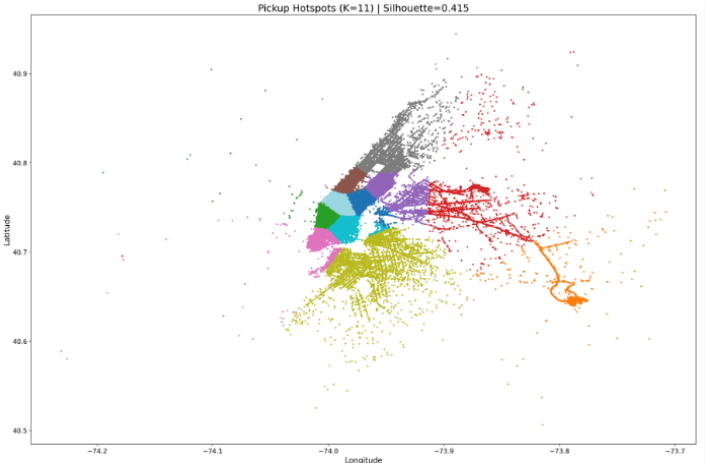
Evaluation

The evaluation of the optimal number of clusters K reveals a trade-off between mathematical simplicity and operational utility. The WCSS plot shows a significant elbow between K=4 and K=6. Beyond this range, the rate of variance captured decreases, suggesting that K=5 (WCSS=251.31) is the best for broad regional planning. Silhouette score peaks sharply at K=2 (0.764) While this indicates the cleanest mathematical separation, a two-cluster model is too generalized for a complex city. Silhouette score fails because it prefers perfectly isolated data groups, whereas taxi data represents a continuous flow of people across the city with no clear gaps between neighborhoods.

For real-world taxi operations, it is best to use a multi-resolution approach where K = 2 to 5 serves as a global model best for big-picture city planning and defining major district boundaries, while K = 8 to 12 acts as a local model best for daily operations



K=5 (Better Global Model for overall city)



K=11(Better for local demand region classification)

Interpretation & Patterns Observed

- **The Central High Density** The small tightly packed clusters in the center represent the Central Business District . This area is characterized by short hop trips and extreme pickup density. These are likely commercial and tourist hotspots where demand is constant throughout the day.
- **The Transit Corridors** Elongated stretching clusters represent Transit Arteries. These follow major highways and bridges. These patterns represent Airport Runs or Suburb to City commutes characterized by longer distances, higher fares and directional flow.
- **The Sparse Periphery** The fragmented clusters on the edges represent Residential Industrial Zones. Demand here is sporadic.

Actionable Recommendations

- To reduce congestion in high density central clusters the city should create fixed taxi pickup and drop zones.K values like 4 or 5 show strong central fragmentation which needs structured handling rather than free movement.
- Launch Corridor Shuttles for Transit Spokes:At higher K values like K = 11, these corridor patterns become more visible and consistent. Introduce High Frequency Shuttle Vans or Bus Rapid Transit services along these routes during peak hours to reduce long distance taxi dependency.
- Incentivize Last Mile Coverage in Peripheral Zones:The sparse outer clusters show low demand and weak driver presence leading to accessibility issues. Action Implement Zone Based Subsidies or fare incentives for trips originating from peripheral areas to ensure continuous driver availability and balanced citywide coverage.

Task 3: Trip Duration Prediction

Evaluation

The final models were evaluated using Linear Regression (with and without PCA) and Random Forest Regressor with hyperparameter tuning to predict trip duration.Principal Component Analysis (PCA) was applied using 14 components capturing approximately 96.10 percent of variance. However, PCA did not improve model performance. Therefore, the final decision was to use the original feature space for better interpretability and slightly better predictive accuracy.

Linear Regression

- MAE : 518.7980
- RMSE : 2931.8325
- R2 : 0.1438

Random Forest Regressor

Best performing model:

- n_estimators: 200
- max_depth: 20
- min_samples_split: 10
- min_samples_leaf: 10

- MAE : 207.21098692556401
- RMSE: 1002.4910907582794
- R2 : 0.8983708418952255

Final Model Selection

The Random Forest Regressor significantly outperforms Linear Regression models. It achieves:

- Much lower error values (MAE and RMSE)
- Very high explanatory power ($R^2 \approx 0.9$)

This indicates that trip duration has strong non-linear relationships with input features, which Linear Regression fails to capture effectively.

Interpretation and Patterns Observed

- Trip duration is better explained by non-linear relationships rather than linear trends
- Linear models show low explanatory power ($R^2 \approx 0.14$), indicating limited linear dependency in the dataset
- Trip duration shows a clear time-of-day pattern, with lowest values in early morning hours and highest values in mid afternoon, indicating strong congestion effects during daytime
- Weekdays generally have higher average trip duration than weekends, showing consistently higher delays during working days
- Mid-week days show higher trip durations compared to the start and end of the week, indicating uneven traffic
- Trip duration increases during peak daytime hours and decreases during off-peak hours, showing strong variation based on traffic conditions

- There are some extreme long duration trips in the dataset, which act as outliers and affect overall duration values

Actionable Recommendations

- The final Random Forest model can be used by city transport authorities for accurate travel time estimation across urban routes, improving public travel planning and enabling more reliable prediction of trip durations under varying traffic conditions.
- It can support adaptive traffic management strategies by helping authorities adjust signal timings, manage congestion hotspots, and implement time-based traffic control during peak demand periods.
- Transport departments can use the model to improve fleet and route allocation for public transport systems by anticipating longer travel times in advance and optimizing service coverage across different city zones.

Failure Analysis and Limitations

Task 1: Peak-Time Traffic Management

The model shows a failure rate of approximately 30 percent, indicating that nearly one-third of congestion predictions are incorrect. This suggests that the model is not uniformly reliable across all traffic conditions, and its performance varies depending on temporal and contextual factors. Higher error rates are observed during morning peak hours (around 7 to 10 AM) and evening peak hours (around 6 to 8 PM), where traffic conditions are highly dynamic and congestion patterns change rapidly. In contrast, early morning hours show lower error rates, indicating better model performance under stable, low-traffic conditions. Across days of the week, errors are relatively higher during weekends and late-week days, suggesting that irregular travel behavior is more difficult for the model to capture compared to structured mid-week commuting patterns. The model also shows a significant number of missed congestion cases (27210 instances), indicating that it tends to under-detect congestion in borderline or rapidly changing traffic situations, especially during peak transition periods. Overall, the failure analysis shows that prediction errors are concentrated in high-variability traffic conditions, while stable traffic scenarios are predicted more reliably.

Task 2: Spatial Demand Clustering

The clustering results show instability across different values of K, indicating that taxi demand is spatially continuous rather than naturally divided into clearly separable groups. The WCSS (Elbow) curve shows a gradual decrease from K = 2 to around K = 4 to 6, after which improvements become marginal, indicating that no strong natural cluster boundary exists. Although WCSS suggests a weak elbow region around K = 4 to 6, the reduction in within-cluster variance beyond this point is minimal. This indicates that increasing K mainly splits existing dense regions rather than discovering new meaningful spatial structures. Silhouette scores further support this limitation, as the highest value occurs at K = 2, but this represents an overly simplified city structure that does not reflect real-world demand complexity. At higher K values (K = 8 to 12), clusters become fragmented in dense urban areas, where WCSS continues to reduce slightly but does not correspond to meaningful geographic separation. This leads to over-segmentation of the same spatial regions. In peripheral zones, clustering performance weakens further, with clusters forming around sparse and irregular points rather than stable demand regions, indicating weak structural patterns in low-density areas. Overall, both WCSS and silhouette analysis show a trade-off between oversimplification at low K and over-segmentation at high K, confirming that KMeans does not produce a single optimal clustering solution for this dataset.

Task 3: Trip Duration Prediction

The final Random Forest model shows a failure rate of around 0.29, meaning nearly one-third of predictions are inaccurate. This indicates that while the model captures general trip duration patterns, it struggles under complex and highly variable conditions. Errors are higher during late-night and peak daytime hours, where traffic conditions change rapidly, while early morning hours show lower error, reflecting better performance in stable traffic conditions. Across days of the week, errors are slightly higher during mid-week and late-week periods, suggesting that irregular travel behavior reduces prediction accuracy compared to more stable periods. The model also shows increasing error in later months, indicating that seasonal variation and changing travel demand patterns affect performance. Long-distance trips show significantly higher error compared to short trips, highlighting difficulty in handling route variability and traffic uncertainty over longer journeys. Similarly, higher passenger count trips show more deviation, indicating added complexity in group travel scenarios. Weather extremes such as heavy rainfall and snowfall conditions also increase prediction errors, showing limited model robustness under rare environmental conditions. Overall, the model fails most in high-variability situations such as peak hours, long trips, extreme weather, and irregular travel patterns, while performing better in stable and routine conditions.

Future Work

The current study can be extended in multiple directions to improve prediction accuracy, spatial understanding, and real-world applicability of the models. One key improvement is the use of more advanced data mining techniques, or neural network based approaches. In addition, incorporating real-time traffic information, road network structure, and live congestion feeds can significantly enhance performance, especially during peak hours and irregular travel conditions. The congestion model can also be improved by integrating external factors such as special events, public holidays, and accident data, which currently contribute to unexplained prediction errors.

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